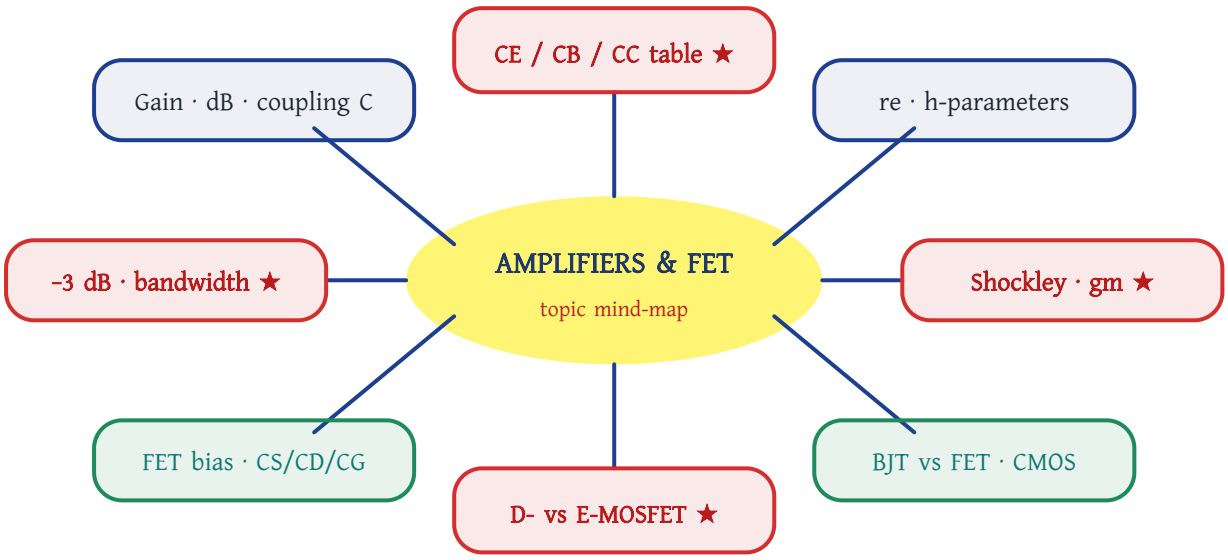
Single-Stage Amplifiers & FET

CE · CB · CC · re & h-models · dB & bandwidth · JFET · MOSFET · FET bias & amplifiers: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = applied / comparison

Syllabus coverage

"**Single stage amplifier**: voltage and current gain, input and output impedance of a common base, common emitter and common collector, h-parameter; **Field effect transistor (FET)**: JFET structure, operation and characteristics. **MOSFET construction, operation and characteristics.**" BJT construction, biasing and the switch are in the BJT notes (lesson 2); diodes in lesson 1.

High-yield list

① CE 180°, CB/CC 0°
② CC: buffer, $A_v \approx 1$
③ $r_e = 26\ mV/I_E$
④ -3 dB = ½ power
⑤ Shockley & g_m
⑥ E-MOSFET: OFF at 0

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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1 The Single-Stage Amplifier

Definition: A single-stage amplifier uses **one** transistor (BJT or FET), biased in its **active region** (BJT) or **saturation / pinch-off region** (FET), to make a small AC signal larger. The DC bias sets the Q-point; the AC signal rides on it. Energy comes from the DC supply, not from the transistor.

Gain	Definition
Voltage A_v	v_o / v_i
Current A_i	i_o / i_i = $-A_v Z_i/R_L$
Power A_p	$A_v \times A_i$
Input Z_i	v_i / i_i (looking in)
Output Z_o	v_o/i_o with $v_s = 0$

Decibels (dB):

Power: $G = 10 \log(P_o/P_i)$

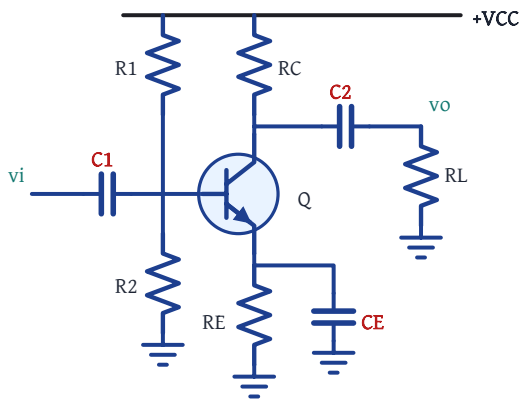
Voltage: $G = 20 \log(V_o/V_i)$

Cascade: **dB values add** (gains multiply)

Voltage ratio	dB	Power ratio
0.707	−3	½
2	+6	4
10 / 100 / 1000	20 / 40 / 60	10^2 … 10^6

Exam trap: A power ratio of 1000 is **30 dB** (10 log), but a voltage ratio of 1000 is **60 dB** (20 log). The bel honours A. G. Bell; 1 B = 10 dB.

2 Coupling & Bypass Capacitors · DC vs AC Analysis



voltage-divider CE amplifier: C1, C2 couple · CE bypasses RE

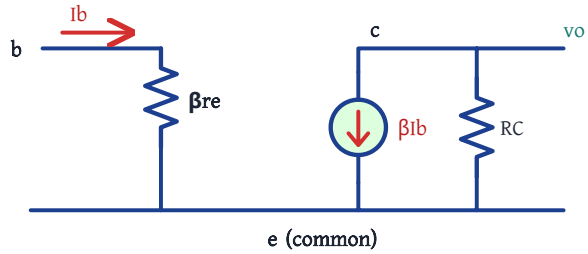
- **Coupling C (C1, C2): pass AC, block DC**, so the source and load do not upset the Q-point. Rule: $X_C \le R/10$ at the lowest signal frequency.
- **Bypass C (C_E):** shorts R_E for AC only. R_E keeps the **DC** stability, yet the AC gain rises from $\approx -R_C/R_E$ to $-R_C/r_e$.

1. **DC analysis:** capacitors = **open**; find I_E, V_{CE} (bias, lesson 2)
2. $r_e = 26\ \text{mV} / I_E$
3. **AC analysis:** capacitors = **short**, DC supply = **AC ground**
4. Replace the transistor by its model; find Z_i, Z_o, A_v, A_i

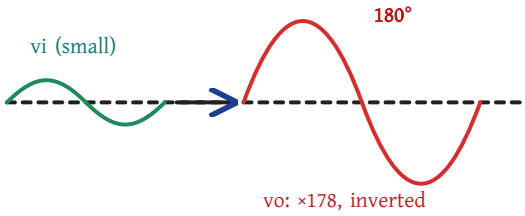
Memory trick: "**DC: caps out, AC: caps in, V_{CC} down**". For AC, R1 and R2 both end on ground, so they appear **in parallel**: R′ = R1 ∥ R2.

3 The r_e Model & the Common-Emitter Stage

r_e model: The forward-biased BE junction has AC resistance **r_e = 26 mV / I_E** (room temperature, $V_T \approx 26$ mV). Seen from the base it looks **β times larger**: $Z_b = \beta r_e$ (= h_{ie}, also called r_π). The output is a current source βI_b (= I_c), with $r_o = 1/h_{oe}$ in parallel.



CE in the r_e model (r_o ignored, C_E bypassing)



CE output is inverted: base ↑ → I_C ↑ → drop on R_C ↑ → V_C ↓

CE, voltage-divider bias	Z_i	Z_o	A_v
R_E bypassed	R1 ∥ R2 ∥ βr_e	R_C (∥ r_o)	$-R_C/r_e$
R_E unbypassed	R1 ∥ R2 ∥ $\beta(r_e + R_E)$	R_C	$-R_C/(r_e + R_E) \approx -R_C/R_E$
With load R_L	same	R_C	replace R_C by R_C ∥ R_L

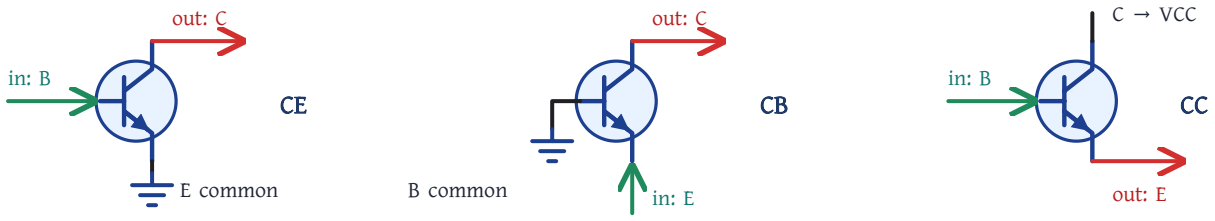
Worked example ✍️: V_{CC} = 12 V, R1 = 47 kΩ, R2 = 10 kΩ, R_C = 3.3 kΩ, R_E = 1 kΩ, β = 100, no load.

1. βR_E = 100 kΩ ≥ 10R2 ✓ → V_B = 12 × 10/57 = 2.11 V, V_E = 1.41 V, I_E = **1.41 mA**
2. r_e = 26/1.41 = **18.5 Ω**; βr_e = 1.85 kΩ; R1 ∥ R2 = 8.25 kΩ
3. Bypassed: Z_i = 8.25k ∥ 1.85k = **1.51 kΩ**, Z_o = 3.3 kΩ, A_v = −3300/18.5 = **−178** (45 dB)
4. C_E removed: Z_b = 100 × 1018.5 Ω = 101.9 kΩ → Z_i = **7.63 kΩ**, A_v = −3300/1018.5 = **−3.24**

Memory trick: CE gain = "**what's in the collector ÷ what's in the emitter**", with a minus sign. Bypassed: only r_e is left in the emitter.

Exam trap: Removing C_E **lowers** the gain (here −178 → −3.24) but **raises** Z_i and makes the gain stable and nearly independent of β (negative feedback).

4 Common Base & Common Collector



the common terminal is shared by input and output (it is AC ground; V_{CC} is AC ground too)

Common base (CB): Input at E, output at C.

$Z_i = R_E \parallel r_e \approx$ **r_e (very low, ~10–50 Ω)**

$Z_o \approx R_C \cdot A_v =$ **$+R_C/r_e$** (no inversion)

$A_i = \alpha \approx$ **0.98–0.99 (< 1)**

Use: RF / high-frequency stages (no Miller effect), low-Z sources.

Common collector (CC) = emitter follower:

Input at B, output at E.

$Z_i = R_B \parallel \beta(r_e + R_E) \rightarrow$ **high**

$Z_o = R_E \parallel r_e \approx r_e \rightarrow$ **very low**

$A_v = R_E/(R_E + r_e) \approx$ **1 (slightly < 1)**

$A_i \approx \beta + 1$. Use: **buffer**, impedance matching.

The master comparison table (memorise)

Property	CE	CB	CC (emitter follower)
Z_i	medium (βr_e, ~1 kΩ)	lowest ($\approx r_e$)	highest (~100 kΩ)
Z_o	medium ($\approx R_C$)	**highest** (device ~MΩ)	**lowest** ($\approx r_e$)
A_v	high, $-R_C/r_e$	high, $+R_C/r_e$	≈ 1
A_i	high (β)	$\alpha < 1$	high ($\beta + 1$)
A_p	highest	moderate	moderate ($\approx A_i$)
Phase	180°	0°	0°
Use	voltage amplifier	RF, current buffer	buffer, matching

Worked example ✍️: Emitter follower, $I_E = 2$ mA, $R_E = 2.2$ kΩ: $r_e = 26/2 = 13$ Ω; $A_v = 2200/2213 =$ **0.994**; $Z_o = 2.2\text{k} \parallel 13 =$ **12.9 Ω**. A weak source now sees a high Z_i, and the load sees a ~13 Ω source.

Memory trick: "**CB: Big output, Baby input**" · "**CC: Copy Cat**": the emitter copies the base voltage (minus 0.7 V), so $A_v \approx 1$.

Exam trap: CB has $A_i < 1$ but a **high** A_v; CC has $A_v < 1$ but a **high** A_i. Only CE inverts. Darlington = two CC stages: $\beta_D \approx \beta_1\beta_2$.

5 h-Parameters (Hybrid Model) in Amplifier Work

Two-port equations (inputs i_1, v_2):

$\mathbf{v_1 = h_{11}i_1 + h_{12}v_2}$

$\mathbf{i_2 = h_{21}i_1 + h_{22}v_2}$

"Hybrid": **mixed units** (Ω, –, –, S). 2nd subscript = configuration: h_{ie}, h_{ib}, h_{ic}.

h	Name	Unit	Typical CE
h_{ie}	input impedance	Ω	~1 kΩ
h_{re}	reverse voltage ratio	–	$\sim 10^{-4}$
h_{fe}	forward current gain	–	50–200
h_{oe}	output admittance	S	~25 µS

Approximate CE (h_{re}, h_{oe} ≈ 0):

$Z_i = h_{ie} \cdot A_i = h_{fe} \cdot Z_o = 1/h_{oe}$

$\mathbf{A_v = -h_{fe}R_L/h_{ie}}$

h_{fe} = 50, h_{ie} = 1.1 kΩ, R_L = 2.2 kΩ → $\mathbf{A_v = -100}$

Link to the r_e model:

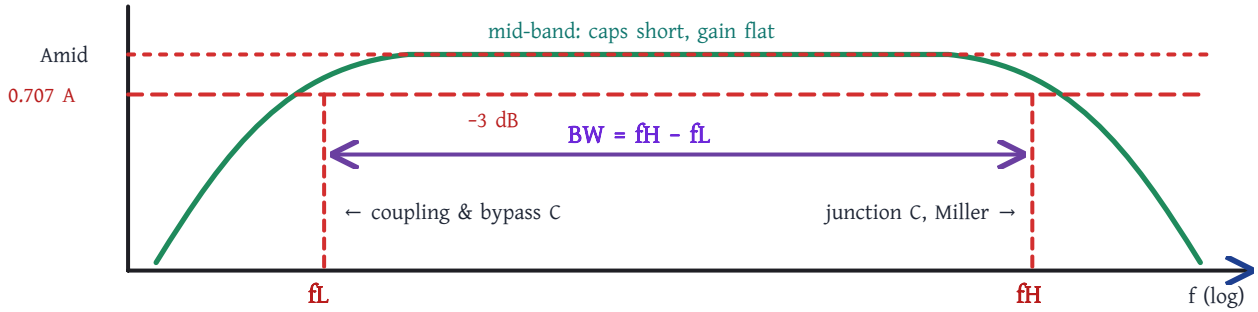
$\mathbf{h_{ie} = \beta r_e} \cdot h_{fe} = \beta$

$h_{ib} = r_e \cdot h_{fb} = -\alpha$

$h_{fc} = -(1 + \beta) \cdot h_{oe} = 1/r_o$

Exam trap: h_{fe} (small-signal AC) ≈ h_{FE} (= β_{DC}, capitals) but they are not identical. h_{fb} is **negative** (≈ -0.99).

6 Frequency Response, -3 dB Points & Bandwidth



RC-coupled stage: gain falls at both ends; each RC edge rolls off at 20 dB/decade (6 dB/octave)

- **Low end f_L:** X_C of C1, C2, C_E grows; $f_L = 1/[2\pi(R_s + Z_i)C1]$. C_E usually sets the highest (dominant) f_L.
- **High end f_H:** C_{be}, C_{bc}, stray C. **Miller effect:** $C_{Mi} = (1 + |A_v|)C_{bc}$.
- At f_L, f_H: gain = **0.707 A_{mid}** (-3 dB), power = ½.

Worked examples ✍️:

① Sheet-3 stage, C1 = 10 µF, R_s ≈ 0: $f_L = 1/(2\pi \times 1511 \times 10^{-5})$ = **10.5 Hz**

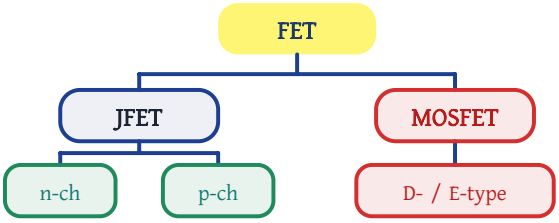
② C_{bc} = 4 pF, A_v = -178: C_{Mi} = 179 × 4 = **716 pF**

③ f_L = 50 Hz, f_H = 20 kHz → BW = **19.95 kHz**

Memory trick: "**Big caps kill the bass, small caps kill the treble**": external µF capacitors set f_L, internal pF capacitances set f_H. Gain × bandwidth ≈ constant.

7 The Field-Effect Transistor (FET) Family

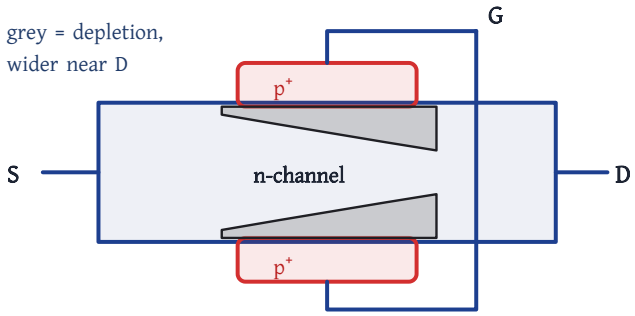
Definition: A FET is a **voltage-controlled**, **unipolar** device: V_{GS} sets the width (or existence) of a conducting **channel** between drain and source; only majority carriers flow (electrons in n-channel). The gate draws almost no current: **$I_G \approx 0$, Z_i very high.**



MOSFET = IGFET; every type comes as n- or p-channel

FET	BJT analogue
Gate G · Drain D · Source S	Base · Collector · Emitter
V_{GS} controls I_D	I_B controls I_C

8 JFET: Construction & Operation



electrons S → D · conventional ID D → S

n-channel JFET: p^+ gates tied together; the gate–channel junction is **reverse-biased**

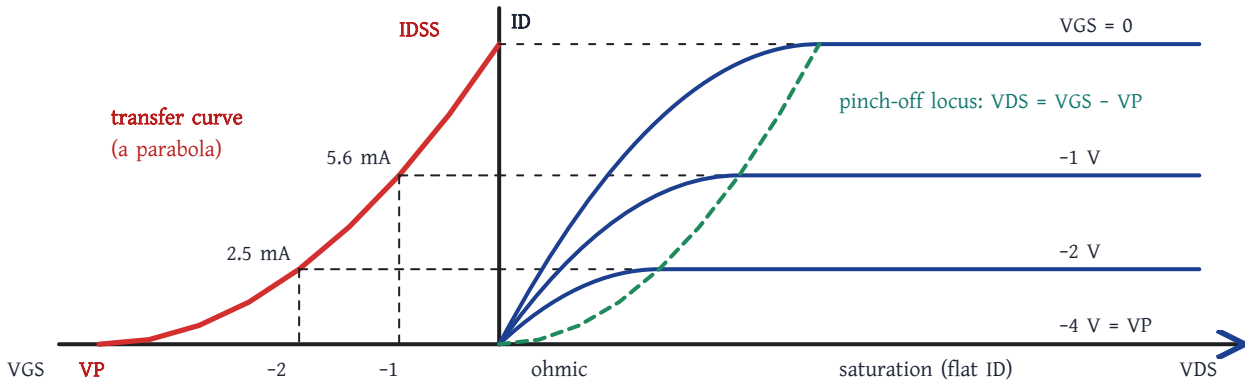
1. $V_{GS} = 0$, small V_{DS}: channel open, I_D rises linearly (**ohmic** region: a voltage-controlled resistor)
2. V_{DS} grows: depletion layers meet near the drain = **pinch-off**; I_D levels off at **I_{DSS}** (max, $V_{GS} = 0$)
3. V_{GS} more negative: channel narrower, I_D smaller
4. $V_{GS} = \mathbf{V_P}$ (= $V_{GS(off)}$): channel closed, $I_D \approx 0$ (**cut-off**)

Region (n-channel, $V_P < 0$)	Condition	I_D
Cut-off	$V_{GS} \le V_P$	≈ 0
Ohmic (triode)	$V_{DS} < V_{GS} - V_P$	$\propto V_{DS}$; r_d set by V_{GS}
Saturation (pinch-off) ← amplifier	$V_{DS} \ge V_{GS} - V_P$	Shockley, ~flat in V_{DS}

Memory trick: JFET = a **garden hose**: the gate squeezes it; squeeze fully ($V_{GS} = V_P$) and the flow stops. Open at rest → **normally ON**, depletion-mode only.

Exam trap: n-channel: $V_{GS} \le 0$; p-channel: $V_{GS} \ge 0$. A forward-biased gate loses the high Z_i. Carriers enter at the **source**, leave at the **drain**.

9 JFET Characteristics & Shockley's Equation



left: transfer curve I_D vs V_{GS} · right: drain curves · I_{DSS} = 10 mA, V_P = −4 V (drawn to scale)

Shockley's equation (saturation): $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ · inverse: $V_{GS} = V_P(1 - \sqrt{(I_D/I_{DSS})})$ · square law → **parabolic** transfer curve

Worked examples ✍️:

① I_{DSS} = 10 mA, V_P = −4 V, V_{GS} = −2 V: $I_D = 10(1 - 0.5)^2$ = **2.5 mA**

② I_{DSS} = 8 mA, V_P = −4 V, V_{GS} = −1 V: $I_D = 8(0.75)^2$ = **4.5 mA**

③ Same FET, I_D = 4.5 mA → $V_{GS} = -4(1 - 0.75)$ = **−1 V** ✓

Memory trick: 4-point sketch: V_{GS} = 0 → I_{DSS} · **0.3V_P** → **$I_{DSS}/2$** · **$V_P/2$** → **$I_{DSS}/4$** · V_P → 0. At V_{GS} = 0, saturation starts at $V_{DS} = |V_P|$.

10 Transconductance g_m

$g_m = \Delta I_D/\Delta V_{GS}$: at constant V_{DS}: the **slope** of the transfer curve, in siemens (mS).

$g_{m0} = 2I_{DSS}/|V_P|$

$g_m = g_{m0}(1 - V_{GS}/V_P) = g_{m0}\sqrt{(I_D/I_{DSS})}$

Amplification factor $\mu = g_m r_d$

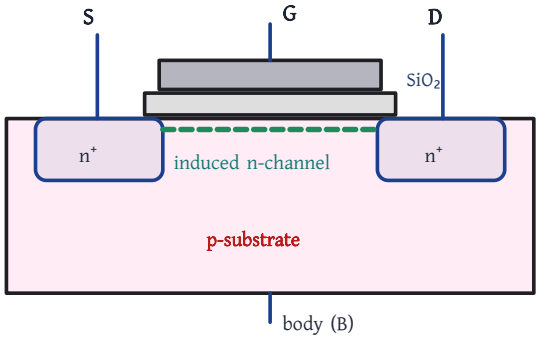
Worked example ✍️: I_{DSS} = 10 mA, V_P = −4 V: $g_{m0} = 20/4$ = **5 mS**; at V_{GS} = −2 V: $g_m = 5(1 - 0.5)$ = **2.5 mS**. g_m is largest at V_{GS} = 0 and zero at V_P.

Exam trap: g_m is the **slope** $\Delta I_D/\Delta V_{GS}$, not I_D/V_{GS}. A BJT at 1 mA has $g_m = I_C/V_T \approx 38$ mS, far above a FET's few mS: **FET stages give lower voltage gain.**

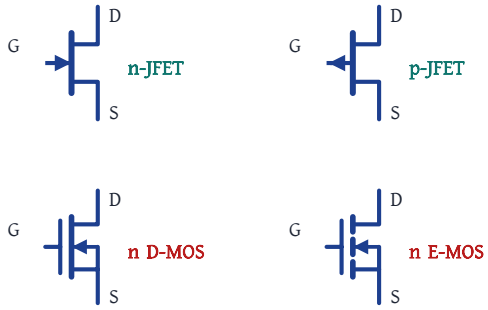
Practical link: In the ohmic region a JFET is a **voltage-controlled resistor** (AGC circuits, analog switches, choppers). Its tiny gate current makes it the input stage of JFET-input op-amps (e.g. TL071) and of pH-meter / electrometer probes.

11 MOSFET: Construction & the Two Types

Definition: Metal-Oxide-Semiconductor FET: the gate is separated from the channel by a thin **SiO_2 insulating layer**, so it acts as a capacitor plate: **$Z_i \approx 10^{10}$–10^{15} Ω** (JFET ~10^8–10^{10} Ω). Four terminals: G, D, S and **substrate (body)**, usually tied to S. Also called **IGFET**.



n-E-MOSFET: $V_{GS} > V_T$ induces the channel



arrow **in** = n-channel · broken = enhancement

	D-MOSFET	E-MOSFET
Channel	diffused in (built-in)	**induced** by $V_{GS} > V_T$
At $V_{GS} = 0$	ON, $I_D = I_{DSS}$	OFF
Modes (n-ch)	depletion (–) **and** enhancement (+)	enhancement only
Equation	Shockley (V_{GS} may be +)	$I_D = k(V_{GS} - V_T)^2$

E-MOSFET (n-channel):

Cut-off: $V_{GS} < \mathbf{V_T}$ **(threshold, $V_{GS(th)}$)**

Triode ($V_{DS} < V_{GS} - V_T$):

$I_D = k[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$

Saturation: $\mathbf{I_D = k(V_{GS} - V_T)^2}$

$g_m = 2k(V_{GS} - V_T)$

Worked example ✍️: $I_{D(on)}$ = 10 mA at $V_{GS(on)}$ = 8 V, V_T = 3 V.

$k = 10/(8 - 3)^2 =$ **0.4 mA/V²**

V_{GS} = 6 V: $I_D = 0.4 \times 3^2 =$ **3.6 mA**; $g_m = 2 \times 0.4 \times 3 =$ **2.4 mS**

V_{GS} = 2 V (< V_T): $I_D =$ **0**

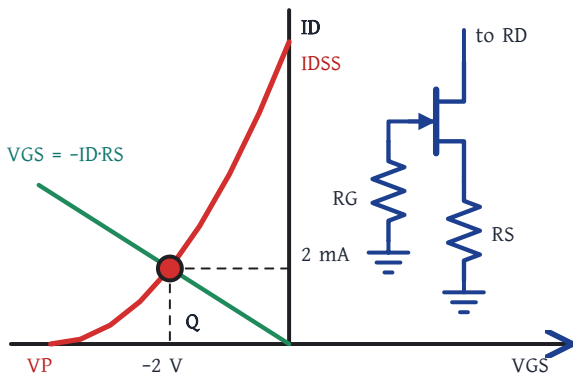
Exam trap: JFET: depletion only; **D-MOSFET: both modes**; E-MOSFET: enhancement only. Thin oxide: **ESD** kills it.

Memory trick: "**E = Empty**": no channel until you **E**nhance past V_T; broken line = empty channel. "**D = Done**": channel already made, normally ON.

IT link: CMOS: **CMOS inverter** = p-channel E-MOSFET (to V_{DD}) over n-channel E-MOSFET, gates and drains tied: in 0 → out 1, in 1 → out 0. One is always OFF, so **static power ≈ 0** (dynamic ≈ CV_{DD}^2f): CPUs and memories are CMOS.

12 FET Biasing

Method	Bias equation	Notes
Fixed bias	$V_{GS} = -V_{GG}$ ($I_G = 0$)	2nd supply; Q varies with device
Self bias	**$V_{GS} = -I_DR_S$**	one supply; JFET, D-MOSFET
Voltage divider	$V_{GS} = V_{DD}R2/(R1 + R2) - I_DR_S$	**most stable**; any FET
Drain feedback	$V_{GS} = V_{DS}$ (R_G from D to G)	E-MOSFET



self bias, solved graphically: the bias line through the origin cuts the parabola at Q

Worked example: self bias ✍️: I_{DSS} = 8 mA, V_P = -4 V, R_S = 1 kΩ, R_D = 3.3 kΩ, V_{DD} = 16 V.

1. $V_{GS} = -I_D$ (I_D in mA) → $I_D = 8(1 - I_D/4)^2$
2. $I_D^2 - 10I_D + 16 = 0$ → I_D = 2 or 8 mA
3. 8 mA gives V_{GS} = -8 V, beyond V_P: reject → **I_D = 2 mA, V_{GS} = -2 V**
4. $V_{DS} = 16 - 2(3.3 + 1) =$ **7.4 V** ≥ $V_{GS} - V_P$ = 2 V ✓

13 FET Amplifiers: CS, CD, CG

Config (BJT twin)	Z_i	Z_o	A_v (r_d ignored)	Phase
Common source (CE)	R_G (MΩ)	R_D	$-g_mR_D$	180°
… R_S unbypassed	R_G	≈ R_D	$-g_mR_D/(1 + g_mR_S)$	180°
Common drain, source follower (CC)	R_G	$R_S \parallel 1/g_m$	$g_mR_S/(1 + g_mR_S) < 1$	0°
Common gate (CB)	$R_S \parallel 1/g_m$	R_D	$+g_mR_D$	0°

Worked example ✍️: Same JFET at I_D = 2 mA: $g_m = (16/4)(1 - 2/4) =$ **2 mS**. C_S bypassed: $A_v = -2 \times 3.3 =$ **-6.6**; unbypassed: -6.6/3 = **-2.2**; source follower (R_S = 1 kΩ): A_v = 2/3 = 0.67, Z_o = 1k ∥ 500 = 333 Ω.

Memory trick: Reuse the BJT table with **S↔E, D↔C, G↔B** and $1/r_e \rightarrow g_m$: CS = CE (inverting gain stage), CD = CC (buffer), CG = CB (low Z_i, RF).

Night-Before-the-Exam Sheet

BJT vs FET (5-mark written favourite)

Feature	BJT	FET
Control · carriers	current (I_B) · bipolar	**voltage** (V_{GS}) · **unipolar**
Input impedance	low–medium (kΩ)	**very high** (10^8–10^{15} Ω)
Gain parameter	β; $g_m = I_C/V_T$ (~38 mS/mA)	$g_m = \Delta I_D/\Delta V_{GS}$ (few mS)
Voltage gain, GBW	**higher**	lower
Noise · thermal	noisier · thermal runaway	quieter · I_D falls with T
Size · static power	larger · more	smaller · CMOS ≈ 0 · ESD-prone

Fact sheet:

$r_e = 26\ mV/I_E$ · $h_{ie} = \beta r_e$

CB: $Z_i \approx r_e$, $A_i = \alpha$

$A_p = A_v A_i$ · 20 log V, 10 log P

$C_{Mi} = (1 + |A_v|)C_{bc}$

$g_{m0} = 2I_{DSS}/|V_P|$

Self bias: $V_{GS} = -I_D R_S$

CE: $A_v = -R_C/r_e$ (bypassed)

CC: $A_v \approx 1$, $Z_o \approx r_e$

−3 dB = 0.707 V = ½ P

$I_D = I_{DSS}(1 - V_{GS}/V_P)^2$

E-MOS: $I_D = k(V_{GS} - V_T)^2$

CS: $A_v = -g_m R_D$

1. Only CE (and CS) invert: 180°
2. Highest power gain: CE
3. Highest Z_i, lowest Z_o: CC
4. Lowest Z_i, $A_i < 1$: CB
5. Bypass C raises A_v
6. JFET ON, E-MOSFET OFF at 0 V
7. D-MOSFET: both modes
8. $V_{GS} = V_P \rightarrow I_D = 0$
9. $V_P/2 \rightarrow I_{DSS}/4$
10. Broken line = E-MOSFET

Top traps: power ×1000 = 30 dB, voltage ×1000 = 60 dB · removing C_E lowers A_v · CB: $A_i < 1$, A_v high · $h_{fb} = -\alpha$ · g_m is a slope · no self bias for E-MOSFET · reject the root with $|V_{GS}| > |V_P|$

All tricks together: DC: caps out, AC: caps in · collector ÷ emitter · Big output, Baby input · Copy Cat · big caps kill the bass · garden hose · 4-point parabola · E = Empty, D = Done · S↔E, D↔C, G↔B

✓ Practice MCQs (exam pattern, 10 marks)

1. Phase shift of a CE amplifier:
(a) 0° (b) 90°
(c) 180° (d) 270°

2. Highest input impedance:
(a) CB (b) CC
(c) CE (d) all equal

3. At I_E = 2 mA, $r_e \approx$
(a) 13 Ω (b) 26 Ω
(c) 52 Ω (d) 1.3 kΩ

4. At f_H the output power falls to:
(a) 70.7% (b) 25%
(c) 0% (d) 50%

5. Which is a unipolar device?
(a) FET (b) BJT
(c) SCR (d) UJT

6. I_{DSS} = 12 mA, V_P = −6 V; I_D at V_{GS} = −3 V:
(a) 6 mA (b) 4 mA
(c) 1.5 mA (d) 3 mA

7. OFF at V_{GS} = 0:
(a) n-JFET (b) D-MOSFET
(c) E-MOSFET (d) p-JFET

8. g_{m0} for I_{DSS} = 12 mA, V_P = −6 V:
(a) 2 mS (b) 4 mS
(c) 6 mS (d) 72 mS

9. Voltage gain 100 in dB:
(a) 20 dB (b) 100 dB
(c) 10 dB (d) 40 dB

10. FET twin of the emitter follower:
(a) CS (b) CD
(c) CG (d) cascode

Answer key → 1-c · 2-b · 3-a · 4-d · 5-a · 6-d · 7-c · 8-b · 9-d · 10-b

Written question: Distinguish JFET, D-MOSFET and E-MOSFET by their transfer characteristics. A JFET has I_{DSS} = 8 mA, V_P = −4 V: find I_D and g_m at V_{GS} = −1 V, and the gain of a CS stage with R_D = 2.2 kΩ (R_S bypassed). 10 marks

1. JFET: $V_P \le V_{GS} \le 0$, $I_D \le I_{DSS}$; D-MOSFET: the same parabola continued into $V_{GS} > 0$; E-MOSFET: I_D = 0 up to V_T, then $k(V_{GS} - V_T)^2$. Sketch all three. 4
2. $I_D = 8(1 - 1/4)^2 = 8 \times 0.5625 =$ **4.5 mA** 2
3. $g_{m0} = 2 \times 8/4 = 4$ mS; $g_m = 4(1 - 0.25) =$ **3 mS** 2
4. $A_v = -g_m R_D = -3 \times 2.2 =$ **−6.6** (180° phase shift) 2

Before the exam ✓

- ✓ Can fill the CE/CB/CC table from memory
- ✓ Can find Z_i, Z_o, A_v with r_e
- ✓ Can convert gains to dB, read −3 dB
- ✓ Can use Shockley and g_m both ways
- ✓ Can draw all four FET symbols

Also practise (written):
① CE with C_E removed
② emitter follower Z_o
③ self bias by quadratic
④ CMOS inverter operation